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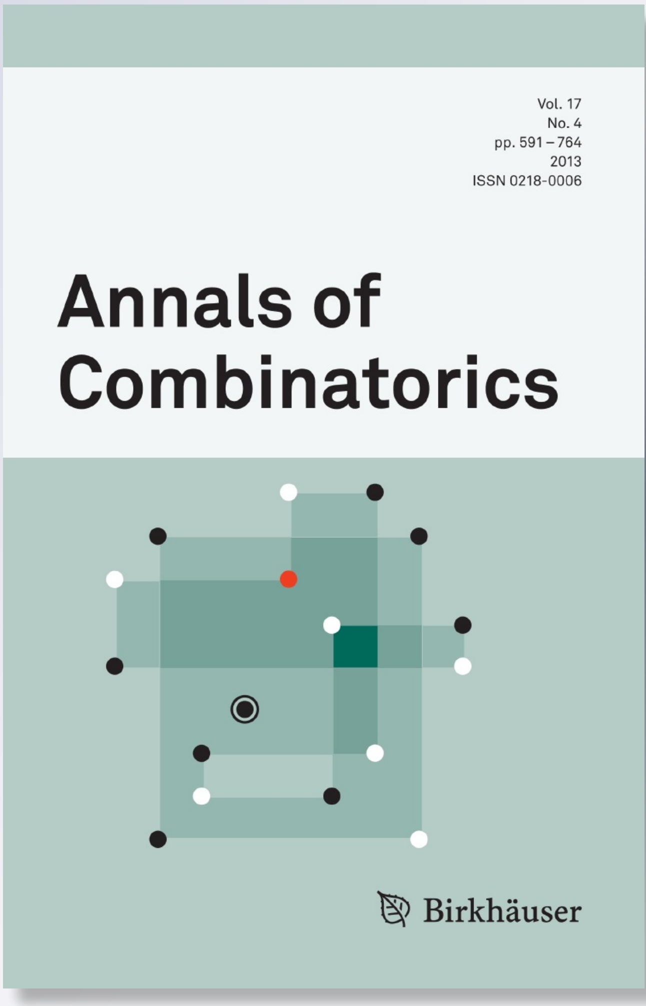

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On Local LYM Identities*

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Abstract. The LYM inequality (Lubell, Yamamoto, Meshalkin) is a generalization of Sperner's theorem for antichains. Kleitman and Harper independently proved that the LYM inequality and the *normalized matching property* (or *local LYM inequality*) are equivalent. Many contributions have been proposed to sharpen the LYM inequality. Noticeably, Ahlswede and Zhang lifted the LYM inequality to an identity, called the *AZ identity*. Thus, one expects that the same sharpening of the local LYM inequality is equivalent to the AZ identity. In this paper, we introduce a *local LYM identity* which sharpens the local LYM inequality and prove that it is equivalent to the AZ identity. The local LYM identity shows local relationships between components in the AZ identity.

Keywords: antichain, AZ identity, local LYM inequality, normalized matching property

1. Introduction

Let $[n] = \{1, 2, \ldots, n\}$, $2^{[n]}$ denote the family of all subsets of $[n]$, $\mathcal{P}_k$ be the set of all subsets of $[n]$ of size k, and $\emptyset$ be the empty set. If $\emptyset \neq \mathcal{F} \subseteq 2^{[n]}$ and $A \not\subset B$ for all $A, B \in \mathcal{F}$, $A \neq B$, then $\mathcal{F}$ is called a *Sperner family* or an *antichain*. The well-known LYM inequality (Lubell, Yamamoto, Meshalkin [8]) is

$$\sum_{F \in \mathcal{F}} \frac{1}{\binom{n}{|F|}} \leq 1 \quad \text{or} \quad \sum_{k=0}^{n} \frac{|\mathcal{F} \cap \mathcal{P}_k|}{\binom{n}{k}} \leq 1,$$

where $\mathcal{F}$ is an arbitrary antichain.

The LYM inequality can be sharpened by adding to the left-hand side non-negative terms or raising its coefficients. In [4], the quantity $\frac{|\mathcal{B}|}{2^n}$ is added to the left-hand side of the LYM inequality, where $\mathcal{B}$ is a family of sets which are incomparable to the sets of $\mathcal{F}$. Erdös et al. proposed to increase terms $\frac{1}{\binom{n}{k}}$ in the LYM inequality [7]. Bey sharpened the LYM inequality by adding suitable terms in polynomial or quadratic

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types of $\frac{|\mathcal{F}\cap\mathcal{P}_k|}{\binom{n}{k}}$ to its left-hand side [2, 3]. Noticeably, Ahlswede and Zhang discovered an elegant identity (called the AZ identity) that can be considered as "the most sharpening" of the LYM inequality [1]. Moreover, the dual types of the AZ identity are also investigated [5, 9].

Additionally, Kleitman and Harper independently proved that the LYM inequality and the *normalized matching property* (or *local LYM inequality*) are equivalent (cf. [8]). As illustrated in Figure 1, it is possible that there exists a sharpening identity of the local LYM inequality. This sharpening identity may be equivalent to the AZ identity. A motivation for this paper is to identify such an identity.

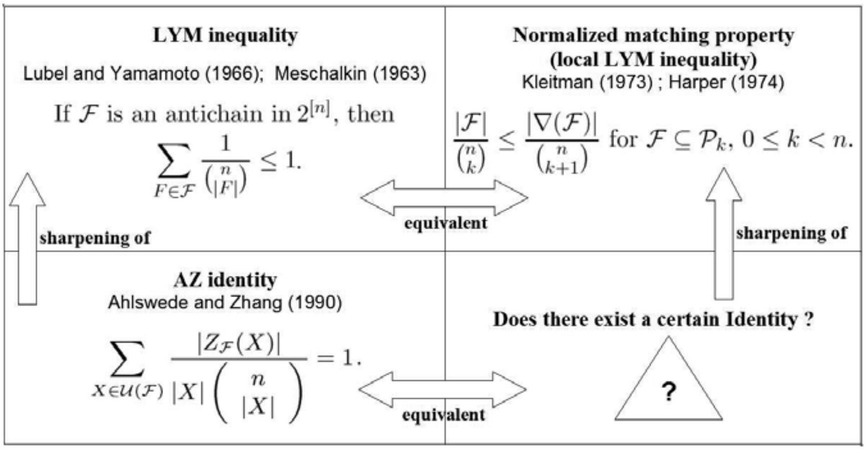

Figure 1: Motivation to find a certain local identity.

In this paper, we present local identities based on the set function defined by Ahlswede and Zhang in [1]. Of these, the main local identity is a sharpening of the local LYM inequality, which also shows relationships among the local LYM inequality, the LYM inequality, and the AZ identity.

The rest of the paper is organized as follows. Section 2 introduces related notations and presents the previous work. Section 3 discusses main theorems of the local identities. Section 4 illustrates consequences and applications of the identities. Finally, some closing remarks are provided in Section 5.

2. Notation and Previous Work

Let $\mathcal{G}$ be the family of $\mathcal{F}$ such that $\emptyset \neq \mathcal{F} \subseteq 2^{[n]}$. For every $\mathcal{F} \in \mathcal{G}$ we define the downset generated by $\mathcal{F}$ as

$$\mathcal{D}(\mathcal{F}) = \{D \subseteq [n] \colon D \subseteq F, \text{ for some } F \in \mathcal{F}\},$$

and the upset generated by $\mathcal{F}$ as

$$\mathcal{U}(\mathcal{F}) = \{U \subseteq [n] \colon U \supseteq F, \text{ for some } F \in \mathcal{F}\},$$

and denote by $\mathcal{F}^c$ the set $\{[n]\backslash F: \ F \in \mathcal{F}\}$. Then for $X \subseteq [n]$, we put

$$Z_{\mathcal{F}}(X) = \begin{cases} \bigcap_{X \supseteq F \in \mathcal{F}} F, & \text{if } X \in \mathcal{U}(\mathcal{F}), \\ \emptyset, & \text{otherwise,} \end{cases}$$

and define $Z^*(\ )$, the dual of $Z(\ )$, as

$$Z^*_{\mathcal{F}}(X) = \begin{cases} \bigcup_{X \subseteq F \in \mathcal{F}} F, & \text{if } X \in \mathcal{D}(\mathcal{F}), \\ [n], & \text{otherwise.} \end{cases}$$

In fact, we only use the value of $Z_{\mathcal{F}}(X)$ whenever $X \in \mathcal{U}(\mathcal{F})$ and the value of $Z^*_{\mathcal{F}}(X)$ whenever $X \in \mathcal{D}(\mathcal{F})$.

Theorem 2.1. ([11]) *Let m be an integer, $\emptyset \notin \mathcal{F} \in \mathcal{G}$. If $|F|+m > 0$ for each $F \in \mathcal{F}$, then*

$$\sum_{X \in \mathcal{U}(\mathcal{F})} \frac{|Z_{\mathcal{F}}(X)|+m}{(|X|+m)\binom{n+m}{|X|+m}} = 1.$$

When $m = 0$ we get the original AZ identity $\sum_{X \in \mathcal{U}(\mathcal{F})} \frac{|Z_{\mathcal{F}}(X)|}{|X|\binom{n}{|X|}} = 1$. If $\mathcal{F}$ is an antichain, $Z_{\mathcal{F}}(F) = F$ for all $F \in \mathcal{F}$. Then,

$$\sum_{F \in \mathcal{F}} \frac{1}{\binom{n}{|F|}} + \sum_{X \in \mathcal{U}(\mathcal{F}) \backslash \mathcal{F}} \frac{|Z_{\mathcal{F}}(X)|}{|X|\binom{n}{|X|}} = 1.$$

This identity implies immediately the LYM inequality because the second sum in the left-hand side is non-negative.

When $m = -1$ we get the identity

$$\sum_{F \in \mathcal{F}} \frac{1}{\binom{n-1}{|F|-1}} + \sum_{X \in \mathcal{U}(\mathcal{F}) \backslash \mathcal{F}} \frac{|Z_{\mathcal{F}}(X)|-1}{(|X|-1)\binom{n-1}{|X|-1}} = 1,$$

for any antichain $\mathcal{F}$, in which the first sum is a useful quantity for many extremal problems of intersecting or Helly families.

For each family $\emptyset \neq \mathcal{A} \subseteq \mathcal{P}_k$, we define the *upper shadow* (or *the shade*) of $\mathcal{A}$ as

$$\nabla \mathcal{A} = \{A \subseteq [n] : \ |A| = k+1, \, A \supset B \text{ for some } \ B \in \mathcal{A}\},$$

and the *lower shadow* (or *the shadow*) of $\mathcal{A}$ as

$$\Delta \mathcal{A} = \{A \subset [n] : \ |A| = k-1, \, A \subset B \text{ for some } \ B \in \mathcal{A}\}.$$

Obviously, if $\mathcal{A} \subseteq \mathcal{P}_k$ then $\nabla \mathcal{A} = \mathcal{U}(\mathcal{A}) \cap \mathcal{P}_{k+1}$ and $\Delta \mathcal{A} = \mathcal{D}(\mathcal{A}) \cap \mathcal{P}_{k-1}$.

The following theorem presents the logical equivalence between the LYM inequality and the normalized matching property (Kleitman, Harper [8]).

Theorem 2.2. (Kleitman, Harper [8]) *The following two inequalities are equivalent.*

(a) *For any antichain* $\mathcal{F} \subseteq 2^{[n]}$*,*

$$\sum_{F\in\mathcal{F}} \frac{1}{\binom{n}{|F|}} \le 1. \tag{2.1}$$

(b) *For* $\mathcal{F} \subseteq \mathcal{P}_k$*,* $0 \le k < n$*,*

$$\frac{|\nabla\mathcal{F}|}{\binom{n}{k+1}} \ge \frac{|\mathcal{F}|}{\binom{n}{k}}. \tag{2.2}$$

3. Main Theorems

The following theorem shows the identity that sharpens the local LYM inequality (2.2) in Theorem 2.2.

Theorem 3.1. (Local LYM identity) *Let k be an integer such that* $1 \le k \le n$*,* $\emptyset \notin \mathcal{A} \in \mathcal{G}$*. Then*

$$\frac{|\mathcal{U}(\mathcal{A})\cap\mathcal{P}_{k-1}|}{\binom{n}{k-1}} + \sum_{X\in\mathcal{U}(\mathcal{A})\cap\mathcal{P}_k} \frac{|Z_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} = \frac{|\mathcal{U}(\mathcal{A})\cap\mathcal{P}_k|}{\binom{n}{k}}.$$

Remark 3.2. Since $|X| = k$ *if* $X \in \mathcal{U}(\mathcal{A})\cap\mathcal{P}_k$*, the identity is equivalent to*

$$\sum_{X\in\mathcal{U}(\mathcal{A})\cap\mathcal{P}_k} |Z_{\mathcal{A}}(X)| = k\,|\mathcal{U}(\mathcal{A})\cap\mathcal{P}_k| - (n-k+1)|\mathcal{U}(\mathcal{A})\cap\mathcal{P}_{k-1}|. \tag{3.1}$$

Our initial proof of Theorem 3.1 is by induction as in [10]. We give here a proof by counting edges of two successive levels of an upset.

Proof of Theorem 3.1. Consider a graph with edges (X,Y) such that $X,Y \in 2^{[n]}$, and $X = Y\cup\{\alpha\}$, $\alpha \notin Y$. We put $\mathcal{U} = \mathcal{U}(\mathcal{A})$ and count in two different ways the number of edges (X,Y) with $X \in \mathcal{U}\cap\mathcal{P}_k$ and $Y \in \mathcal{U}\cap\mathcal{P}_{k-1}$. Since $\mathcal{U}$ is an upset and $Y \in \mathcal{U}$, for all $\alpha \in [n]\backslash Y$ we have $Y\cup\{\alpha\} \in \mathcal{U}$. So the number of the edges is $\sum_{Y\in\mathcal{U}\cap\mathcal{P}_{k-1}} (n-|Y|) = (n-k+1)|\mathcal{U}\cap\mathcal{P}_{k-1}|$. Then we consider the number of edges (X,Y) such that $X \in \mathcal{U}\cap\mathcal{P}_k$ and $Y = X\backslash\{\beta\}$ for $\beta \in X$. The number of these edges is $\sum_{X\in\mathcal{U}\cap\mathcal{P}_k} |X| = k|\mathcal{U}\cap\mathcal{P}_k|$. However, for each $X \in \mathcal{U}\cap\mathcal{P}_k$ and $\beta \in X$, we have two cases for the set $Y = X\backslash\{\beta\}$.

Case 1. $Y \in \mathcal{U}\cap\mathcal{P}_{k-1}$. By the above argument the number of edges in this case is $(n-k+1)|\mathcal{U}\cap\mathcal{P}_{k-1}|$;

Case 2. $Y \notin \mathcal{U}\cap\mathcal{P}_{k-1}$, so (X,Y) belongs to the edges exiting $\mathcal{U}$ from X. For each $X \in \mathcal{U}\cap\mathcal{P}_k$, the number of the edges that leave $\mathcal{U}$ from X is $|Z_{\mathcal{A}}(X)|$ (cf. [1]). Thus the total number of edges in this case is the number of the edges leaving $\mathcal{U}\cap\mathcal{P}_k$, which is $\sum_{X\in\mathcal{U}\cap\mathcal{P}_k} |Z_{\mathcal{A}}(X)|$.

Finally, we deduce

$$k|\mathcal{U}\cap\mathcal{P}_k| = (n-k+1)\,|\mathcal{U}\cap\mathcal{P}_{k-1}| + \sum_{X\in\mathcal{U}\cap\mathcal{P}_k} |Z_{\mathcal{A}}(X)|,$$

which is equivalent to Identity (3.1). ∎

In the same way as in [5], applying Theorem 3.1 with $\mathcal{A}^c$ and $n-k$, we obtain the dual local identity that is given in the following theorem.

Theorem 3.3. (Dual local LYM identity) *Let k be an integer such that $0 \le k < n$, $[n] \notin \mathcal{A} \in \mathcal{G}$. Then*

$$\frac{|\mathcal{D}(\mathcal{A}) \cap \mathcal{P}_{k+1}|}{\binom{n}{k+1}} + \sum_{X \in \mathcal{D}(\mathcal{A}) \cap \mathcal{P}_k} \frac{n - |Z^*_{\mathcal{A}}(X)|}{(n-|X|)\binom{n}{|X|}} = \frac{|\mathcal{D}(\mathcal{A}) \cap \mathcal{P}_k|}{\binom{n}{k}}.$$

Similarly to the case of the LYM inequality and the normalized matching property, the logical equivalence of the AZ identity and the local identity for upsets is given in the following theorem.

Theorem 3.4. (Logical equivalence of the AZ identity and the local identity) *The following identity*

$$\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{|Z_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} = 1, \quad \textit{for } \emptyset \notin \mathcal{A} \in \mathcal{G} \tag{3.2}$$

holds if and only if for each integer k, $1 \le k \le n$, $\emptyset \notin \mathcal{A} \in \mathcal{G}$ we have

$$\frac{|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_{k-1}|}{\binom{n}{k-1}} + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k} \frac{|Z_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} = \frac{|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k|}{\binom{n}{k}}. \tag{3.3}$$

Proof. First, we prove that (3.3) implies (3.2). Using (3.3) we have

$$\begin{aligned}
\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{|Z_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} &= \sum_{k=1}^{n} \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k} \frac{|Z_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} \\
&= \sum_{k=1}^{n} \left(\frac{|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k|}{\binom{n}{k}} - \frac{|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_{k-1}|}{\binom{n}{k-1}} \right) \\
&= \frac{|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_n|}{\binom{n}{n}} - \frac{|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_0|}{\binom{n}{0}},
\end{aligned}$$

which is $1-0=1$. So the AZ identity (3.2) holds.

Now suppose (3.2) is true. Put $W(X) = |X|\binom{n}{|X|}$. Given an integer k, $1 \le k \le n$, we apply the AZ identity to the family $\mathcal{B} = \mathcal{A} \cup \mathcal{P}_k$ to get

$$\sum_{\substack{X \in \mathcal{U}(\mathcal{B}) \\ |X|<k}} \frac{|Z_{\mathcal{B}}(X)|}{W(X)} + \sum_{\substack{X \in \mathcal{U}(\mathcal{B}) \\ |X|=k}} \frac{|Z_{\mathcal{B}}(X)|}{W(X)} + \sum_{\substack{X \in \mathcal{U}(\mathcal{B}) \\ |X|>k}} \frac{|Z_{\mathcal{B}}(X)|}{W(X)} = 1,$$

then consider the set $Z_{\mathcal{B}}(X)$ according to three different cases of the cardinality of X ($|X| < k$, $|X| = k$, and $|X| > k$) for $X \in \mathcal{U}(\mathcal{B})$.

Case 1. $X \in \mathcal{U}(\mathcal{B})$ and $|X| < k$. We have $X \notin \mathcal{U}(\mathcal{P}_k)$ because of $|X| < k$, hence $Z_{\mathcal{B}}(X) = Z_{\mathcal{A}}(X)$;

Case 2. $X \in \mathcal{U}(\mathcal{B})$ and $|X| = k$. So $X \in \mathcal{P}_k$. Hence $Z_{\mathcal{B}}(X)$ is X if $X \notin \mathcal{U}(\mathcal{A})$ and is $Z_{\mathcal{A}}(X)$ if $X \in \mathcal{U}(\mathcal{A})$;

Case 3. $X \in \mathcal{U}(\mathcal{B})$ and $|X| > k$. In this case, X contains all subsets of $[n]$ of size k, which deduces $Z_{\mathcal{B}}(X) \subseteq \bigcap_{B \in \mathcal{P}_k} B = \emptyset$, and hence $Z_{\mathcal{B}}(X) = \emptyset$.

Now we have

$$\sum_{\substack{X \in \mathcal{U}(\mathcal{A}) \\ |X| < k}} \frac{|Z_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} + \sum_{\substack{X \notin \mathcal{U}(\mathcal{A}) \\ X \in \mathcal{P}_k}} \frac{|X|}{|X|\binom{n}{|X|}} + \sum_{\substack{X \in \mathcal{U}(\mathcal{A}) \\ X \in \mathcal{P}_k}} \frac{|Z_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} + 0 = 1,$$

which yields

$$\sum_{\substack{X \in \mathcal{U}(\mathcal{A}) \\ |X| \le k}} \frac{|Z_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} + \frac{|\mathcal{P}_k| - |\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k|}{\binom{n}{k}} = 1.$$

Hence,

$$\sum_{\substack{X \in \mathcal{U}(\mathcal{A}) \\ |X| \le k}} \frac{|Z_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} = \frac{|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k|}{\binom{n}{k}}. \tag{3.4}$$

Replacing k with $k-1$ in (3.4), we have

$$\sum_{\substack{X \in \mathcal{U}(\mathcal{A}) \\ |X| \le k-1}} \frac{|Z_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}} = \frac{|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_{k-1}|}{\binom{n}{k-1}}.$$

Finally, subtracting the sum in Identity (3.4) and the one in the last identity, we get Identity (3.2). ∎

4. Applications of Local Identities

4.1. Sharpening Local LYM Inequalities

We recall a fundamental lemma of Sperner.

Lemma 4.1. (Sperner [6, 8]) *Let* $\emptyset \neq \mathcal{A} \subseteq \mathcal{P}_k$. *Then*

$$\frac{|\nabla \mathcal{A}|}{\binom{n}{k+1}} \ge \frac{|\mathcal{A}|}{\binom{n}{k}}, \quad \textit{if } 0 \le k < n,$$

and

$$\frac{|\Delta \mathcal{A}|}{\binom{n}{k-1}} \ge \frac{|\mathcal{A}|}{\binom{n}{k}}, \quad \textit{if } 0 < k \le n.$$

The inequalities in Lemma 4.1 are also called *local LYM inequalities*, which can be applied to prove the LYM inequality. Now we use the local identities to establish identities corresponding to the local LYM inequalities.

If $\mathcal{A} \subseteq \mathcal{P}_k$ then $\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k = \mathcal{A}$ and $\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_{k+1} = \nabla \mathcal{A}$. Therefore, from Theorem 3.1 (replacing k with $k+1$), we have

Lemma 4.2. *Let* $\emptyset \neq \mathcal{A} \subseteq \mathcal{P}_k$, $0 \le k < n$. *Then*

$$\frac{|\mathcal{A}|}{\binom{n}{k}} + \frac{1}{(k+1)\binom{n}{k+1}} \sum_{X \in \nabla \mathcal{A}} |Z_{\mathcal{A}}(X)| = \frac{|\nabla \mathcal{A}|}{\binom{n}{k+1}}. \tag{4.1}$$

Because the second term of the left-hand side of (4.1) is always non-negative, (4.1) implies immediately the first inequality of Lemma 4.1. Moreover, the equality in the first local LYM inequality happens if and only if $Z_{\mathcal{A}}(X)$ is empty for all $X \in \nabla\mathcal{A}$.

Similarly, Theorem 3.3 yields (replacing k with $k-1$)

Lemma 4.3. *Let* $\emptyset \neq \mathcal{A} \subseteq \mathcal{P}_k$, $0 < k \leq n$. *Then*

$$\frac{|\mathcal{A}|}{\binom{n}{k}} + \frac{\sum_{X\in\Delta\mathcal{A}}(n-|Z^*_{\mathcal{A}}(X)|)}{(n-k+1)\binom{n}{k-1}} = \frac{|\Delta\mathcal{A}|}{\binom{n}{k-1}}. \tag{4.2}$$

Since $Z^*_{\mathcal{A}}(X) \subseteq [n]$, $n-|Z^*_{\mathcal{A}}(X)| \geq 0$, we can deduce from (4.2) the second local LYM inequality in Lemma 4.1. The equality in the second local LYM inequality happens whenever $Z^*_{\mathcal{A}}(X) = [n]$ for all $X \in \Delta\mathcal{A}$.

In fact, the proofs of the LYM inequality in literature, which use Lemma 4.1, contain implicitly *another local inequality*. The following corollary shows explicitly the inequality that is *the local version for the LYM inequality*. This local inequality implies directly the LYM inequality.

Corollary 4.4. *Let $\mathcal{A}$ be an antichain and k be an integer,* $1 \leq k \leq n$. *Then*

$$\frac{|\mathcal{A}\cap\mathcal{P}_k|}{\binom{n}{k}} \leq \frac{|\mathcal{U}(\mathcal{A})\cap\mathcal{P}_k|}{\binom{n}{k}} - \frac{|\mathcal{U}(\mathcal{A})\cap\mathcal{P}_{k-1}|}{\binom{n}{k-1}}.$$

Proof. We write the sum in the left-hand side of the identity of Theorem 3.1 as

$$\sum_{X\in\mathcal{A}\cap\mathcal{P}_k} \frac{1}{\binom{n}{|X|}} + \sum_{X\in(\mathcal{U}(\mathcal{A})\cap\mathcal{P}_k)\setminus\mathcal{A}} \frac{|Z_{\mathcal{A}}(X)|}{|X|\binom{n}{|X|}}.$$

The first sum of this expression is $\frac{|\mathcal{A}\cap\mathcal{P}_k|}{\binom{n}{k}}$, and the second is non-negative. Hence the result holds. ∎

Let r be the maximum size of all sets in the antichain $\mathcal{A}$. We deduce from Corollary 4.4 the inequality $\sum_{A\in\mathcal{A}}\frac{1}{\binom{n}{|A|}} \leq \frac{|\mathcal{U}(\mathcal{A})\cap\mathcal{P}_r|}{\binom{n}{r}}$ to imply the LYM inequality since the right-hand side is less than or equal to 1. If $|A| \leq \frac{n}{2}$ for all $A \in \mathcal{A}$, then

$$\sum_{A\in\mathcal{A}} \frac{1}{\binom{n}{|A|}} \leq \frac{\left|\mathcal{U}(\mathcal{A})\cap\mathcal{P}_{\lfloor\frac{n}{2}\rfloor}\right|}{\binom{n}{\lfloor\frac{n}{2}\rfloor}}.$$

4.2. Parametrised AZ Style Identities

We can deduce the parametrised identity of Theorem 2.1 from the local identities. The following lemma can be derived from Theorem 3.1.

Lemma 4.5. (Parametrised local identity) *Let m, k be integers and assume* $1 \leq k \leq n$, $m+k > 0$, $\emptyset \notin \mathcal{A} \in \mathcal{G}$. *Then*

$$\frac{|\mathcal{U}(\mathcal{A})\cap\mathcal{P}_{k-1}|}{\binom{m+n}{m+k-1}} + \sum_{X\in\mathcal{U}(\mathcal{A})\cap\mathcal{P}_k} \frac{m+|Z_{\mathcal{A}}(X)|}{(m+|X|)\binom{m+n}{m+|X|}} = \frac{|\mathcal{U}(\mathcal{A})\cap\mathcal{P}_k|}{\binom{m+n}{m+k}}.$$

Now we can write the left-hand side of the identity in Theorem 2.1 as

$$\sum_{k=1}^{n} \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k} \frac{m + |Z_{\mathcal{A}}(X)|}{(m+|X|)\binom{m+n}{m+|X|}},$$

and use Lemma 4.5 to deduce easily Theorem 2.1.

Similarly, we can deduce from Theorem 3.3 the parametrised version of the dual local identity as follows.

Lemma 4.6. (Dual parametrised local identity) *Let q, k be integers and assume $0 \leq k < n$, $k < q$, $[n] \notin \mathcal{A} \in \mathcal{G}$. Then*

$$\frac{\left|\mathcal{D}(\mathcal{A}) \cap \mathcal{P}_{k+1}\right|}{\binom{q}{k+1}} + \sum_{X \in \mathcal{D}(\mathcal{A}) \cap \mathcal{P}_k} \frac{q - |Z^*_{\mathcal{A}}(X)|}{(q-|X|)\binom{q}{|X|}} = \frac{|\mathcal{D}(\mathcal{A}) \cap \mathcal{P}_k|}{\binom{q}{k}}.$$

The dual parametrised local identity can be used to prove the dual identity of the identity in [11, Theorem 1].

4.3. Half-Way AZ-Style Identities

In the same method as above, we obtain the parameterised version of the inequality (3.4)

$$\sum_{\substack{X \in \mathcal{U}(\mathcal{A}) \\ |X| \leq k}} \frac{|Z_{\mathcal{A}}(X)| + m}{(|X|+m)\binom{n+m}{|X|+m}} = \frac{|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k|}{\binom{n+m}{k+m}}, \tag{4.3}$$

where m and k are integers such that $1 \leq k \leq n$, $m + k > 0$.

Putting $k = \left\lfloor \frac{n}{2} \right\rfloor$, we have an identity for the poset $\mathcal{P}_{\leq \left\lfloor \frac{n}{2} \right\rfloor} = \bigcup_{k \leq \left\lfloor \frac{n}{2} \right\rfloor} \mathcal{P}_k$, which is the case of the half-way AZ-style identities [12].

5. Closing Remarks

The sum in the original AZ identity is over all $X \subseteq [n]$ because $Z(\ )$ is empty if $X \notin \mathcal{U}(\mathcal{A})$ by the definition of Ahlswede and Zhang. In the first version of the AZ dual identity [5], the sum is also over all $X \subseteq [n]$. Therefore, this version does not show extremal properties. In fact, the sum should be over all $X \in \mathcal{U}(\mathcal{A})$ for the case of usual AZ-style identities, and over all $X \in \mathcal{D}(\mathcal{A})$ for the case of dual identities rather than over all $X \subseteq [n]$.

Usually, empty unions are $\emptyset$ and empty intersections are $[n]$. Notice that the current definitions of $Z(\ )$ and $Z^*(\ )$ do not follow this convention of set theory. Unfortunately, there are some risks for the properties of $Z(\ )$ and $Z^*(\ )$ in trivial cases of $X \notin \mathcal{U}(\mathcal{A})$ or $X \notin \mathcal{D}(\mathcal{A})$. For example, the property $Z_{\mathcal{A}\cup\mathcal{B}}(X) = Z_{\mathcal{A}}(X) \cap Z_{\mathcal{B}}(X)$ holds whenever $X \in \mathcal{U}(\mathcal{A})$ and $X \in \mathcal{U}(\mathcal{B})$ but it is not correct if $X \in \mathcal{U}(\mathcal{A})$, $Z_{\mathcal{A}}(X) \neq \emptyset$ and $X \notin \mathcal{U}(\mathcal{B})$. In order to avoid these risks, we suggest that $Z_{\mathcal{A}}(X)$ is $[n]$ if $X \notin \mathcal{U}(\mathcal{A})$, and $Z^*_{\mathcal{A}}(X)$ is $\emptyset$ if $X \notin \mathcal{D}(\mathcal{A})$. Of course, the summation in the AZ-style identities is always over all $X \in \mathcal{U}(\mathcal{A})$ or $X \in \mathcal{D}(\mathcal{A})$.

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